

Heart rate estimation in PPG signals using Convolutional-Recurrent Regressor

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ABSTRACT

Heart rate monitoring using PPG signal has emerged as an attractive as well as an applied research problem which enjoys a renewed interest in the recent years. Spectral analysis of PPG for heart rate monitoring, though effective when the subject is at rest, suffers from performance degradation in case of motion artifacts which mask the peak related with the actual heart rate. Leveraging the recent advancements in deep (machine) learning and exploiting the signal, spectral and time-frequency perspectives, we introduce an effective method for heart rate estimation from PPG signals acquired from subjects performing different exercises. We extract a set of features characterizing the signal and feed these feature sequences to a hybrid convolutional-recurrent neural network (C-RNN) in a regression framework. Experimental study on the benchmark IEEE signal processing cup dataset reports low error rates reading 2.41 ± 2.90 bpm for subject-dependent and 3.8 ± 2.3 bpm for subject-independent protocol thus, validating the ideas put forward in this study.

1. Introduction

Monitoring of average heart rate (beats per minute or bpm) is one of the oldest yet the most widely used method for diagnosis of cardiovascular diseases (CVDs) [1]. Modalities commonly employed for this purpose include Electrocardiography (ECG), Phonocardiography (PCG) and Photoplethysmography (PPG) as illustrated in Fig. 1. It can be seen from the figure that ECG and PPG have a single distinct peak during a heart cycle while PCG has two peaks referring to the two major sounds which are heard when the heart beats. Traditionally, 12-electrode ECG or PCG signals, collected in noise free environment are the common modalities for diagnosis of CVDs as they are major source of mortality in the world [2]. However, in the recent years, heart rate monitors (HRMs) which can be used in real time have gained notable popularity. Generally, these monitors are based on ECG and PPG technology, for example, Fitbit Charge 3 [3] and Garmin Forerunner 235 [4] are using PPG & ECG technologies. POLAR H10 [5], on the other hand, is based solely on the ECG technology. It must be pointed out these monitors are not only useful for estimating heart rate but they are also being used in other applications like pulse rate tracking, blood pressure measurement, hear

rate variability and others. Moreover, PPG technology is also employed in applications like heart rate variability [6], establishing relationship between heart & respiration rate [7], effect of stimuli on heart rate [8] etc. However, heart rate estimation and monitoring which serve as strong indicators of various CVDs, still remain major applications of the PPG technology.

Estimation of heart rate generally involves observing the biomedical signal for a specific period of time and, calculating the average occurrence of the heart beats. Monitoring, on the other hand, refers to the continuous estimation of average heart rate and, in general, lasts for longer duration in comparison to estimation. This allows detection of a sudden cardiovascular event which may not appear during short duration due to its non cyclo-stationary nature.

Heart rate estimation is considered an accepted modality to detect the occurrence of cyclo-stationary events like instances of continuous murmur, heart sounds (like S1, S2) etc. which in turn allow detection of various CVDs [9–11]. Likewise, continuous monitoring of heart rate is important for identification of rare events like occurrence of S3 and S4 sounds, extra systole, arrhythmias and so on [12]. Among these, heart rate monitoring is particularly important for detection of the

Abbreviations: IEEE Signal Processing Cup, (SPC); Singular Spectrum Analysis, (SSA); Sparse signal reconstruction, (SSR); Focal under-determined system solver, (FOCUSS); Artificial neural networks, (ANN); Convolutional neural networks, (CNN); Long short term memory, (LSTM); Binary Cornet, (b-Cornet); Convolutional and recurrent neural networks, (C-RNN); and fractional Brownian motion, (fBm).

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arrhythmias. Arrhythmias refers to an abnormal heartbeat pattern which usually appears during exercise and, in some cases for individuals, who do not have a specific heart condition but are prone to it. Hence, detection of this irregular heartbeat rhythm through continuous monitoring during exercise can serve as early warning/diagnosis of CVDs [13–17]. From a clinical perspective, since arrhythmias exhibits irregular heart beat during exercise or once the individual is performing some energy intensive physical task, heart rate needs to be monitored in the same scenario. A downside of this, naturally, is the introduction of motion artifacts in the PPG signals which make automatic analysis a challenging problem.

In the recent years, efforts have been made to develop heart rate monitoring systems based on the PPG technology alone. PPG is an inexpensive technology that employs an LED as source and a PD (photo diode) as receiver, hence, it is very light and can easily be accommodated in wearable HRMs. However, PPG signal is heavily affected due to body motion. During motion, signal is deteriorated due to noise which is high in amplitude and low in frequency. These high amplitude low frequency components which have spectral contents similar to heart rate are known as motion artifacts (MAs). Fig. 2 illustrates the effects of motion artifacts on PPG signals which are changed in such a way that none of the peaks in the PPG spectra shows the actual heart rate.

Heart rate estimation from PPG signals has been thoroughly investigated in the literature. Research endeavours to estimate the heart rate from PPG signal can be broadly categorized into signal processing-based methods and machine learning-based methods. Signal processing techniques [19] exploit the signal characteristics for heart rate estimation while machine learning-based techniques employ a set of labeled training samples to learn to calculate the heart rate. Once trained, the model can subsequently be employed to estimate the heart rate from unseen signals [20]. Another important aspect is that signal processing techniques, in most cases, are subject-independent. On the other hand, major proportion of machine learning-based methods are subject-dependent i.e. each subject has its own dedicated model. In the current study, we target a hybrid approach that borrows effective aspects from both signal processing and machine learning paradigms. On one hand, we employ signal filtering and extract a set of features using signal processing-based methods. On the other hand, leveraging the recent advancements in machine learning, we feed the extracted features to an end-to-end convolutional-recurrent network (C-RNN) in the regression framework to estimate the heart rate. Unlike most of the studies which either use signal characteristics or employ convolutional networks to automatically learn features, we first extract features (with an intuitive explanation) from the signal and then feed those to the convolutional layers to learn high level representations. While learning these representations, the convolutional layers also (implicitly) serve as weighting the features as a function of their effectiveness. Furthermore, the proposed network architecture is relatively shallow which means fewer parameters to learn and lesser memory requirements (for

deployment of trained model) at inference time. Not only we benefit from rich signal features and ability of the C-RNN to model heart rate, but the technique is also generalized in the sense that the same method can be employed in both subject-dependent and subject-independent scenarios. The key highlights of this study are outlined in the following.

- Heart rate estimation utilizing a combination of convolutional and recurrent neural networks representing a Convolutional-Recurrent Regressor.
- Investigation of multiple filters to identify the optimal filter(s) to be applied to the signals for pre-processing to restrict the solution space for fast heart rate estimation.
- Learning of robust high level features using 1D convolutions applied on signal, spectral and time-frequency feature sequences.
- Proposition of a shallow neural network allowing the estimation and monitoring of average heart rate in the real time.
- A comprehensive experimental study on publicly available IEEE signal processing cup 2015 dataset validating the effectiveness of the proposed model.

The contents of this paper are organized into seven sections. We first discuss well-known contributions to heart rate estimation from PPG signals in Section 2. Dataset employed in our experimental study is introduced in Section 3 while the technical details of our proposed method are presented in Section 4. This is followed by the experimental protocol and the reported results in Section 5. A detailed discussion is presented in Section 6 while Section 7 concludes the paper with a recall of our key findings.

2. Related work

Analysis of PPG signals has remained an attractive problem for many years, for researchers in signal processing as well as the pattern classification community. Consequently, the problem has been addressed using signal processing as well as machine learning based techniques. We discuss each of these in the following sections.

2.1. Signal processing-based techniques

Research endeavors on the analysis of PPG signals are primarily focused on the use of multi-resolution signal decomposition and signal conditioning. However, adaptive filtering is also used but to a relatively lesser extent. Among signal decomposition-based methods, Fu et al. [21], for instance, compared the performance of wavelet decomposition with mean filtering and concluded that six level wavelet decomposition resulted in the same performance as that of moving average filters of length $F_s/2$, $F_s/4$ & $F_s/8$ for noise removal. In 2014, Z. Zhang [22] introduced an interesting theory for heart rate analysis. The approach relies on band-pass filtering (BPF), signal decomposition using singular

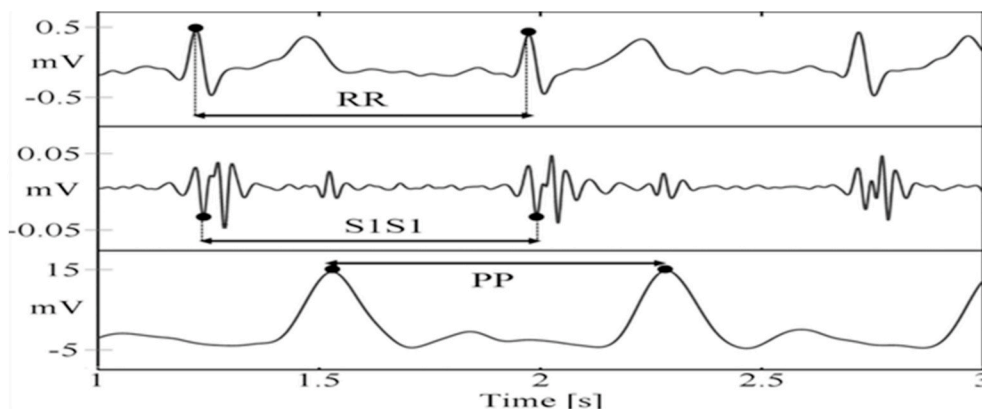


Fig. 1. Common modalities for diagnosis of cardiovascular diseases. Top: Electrocardiogram (ECG), Middle: Phonocardiogram (PCG), Bottom: Photoplethysmogram (PPG). ECG, PCG and PPG record the electrical current flow, heart sounds generated and blood flow during heart beat, respectively [18]. ‘RR’ is the separation between major peaks in QRS complex in the ECG, ‘S1S1’ is the duration between major heart sounds namely ‘S1’ in the PCG signals and ‘PP’ is the peak to peak separation in the PPG signal.

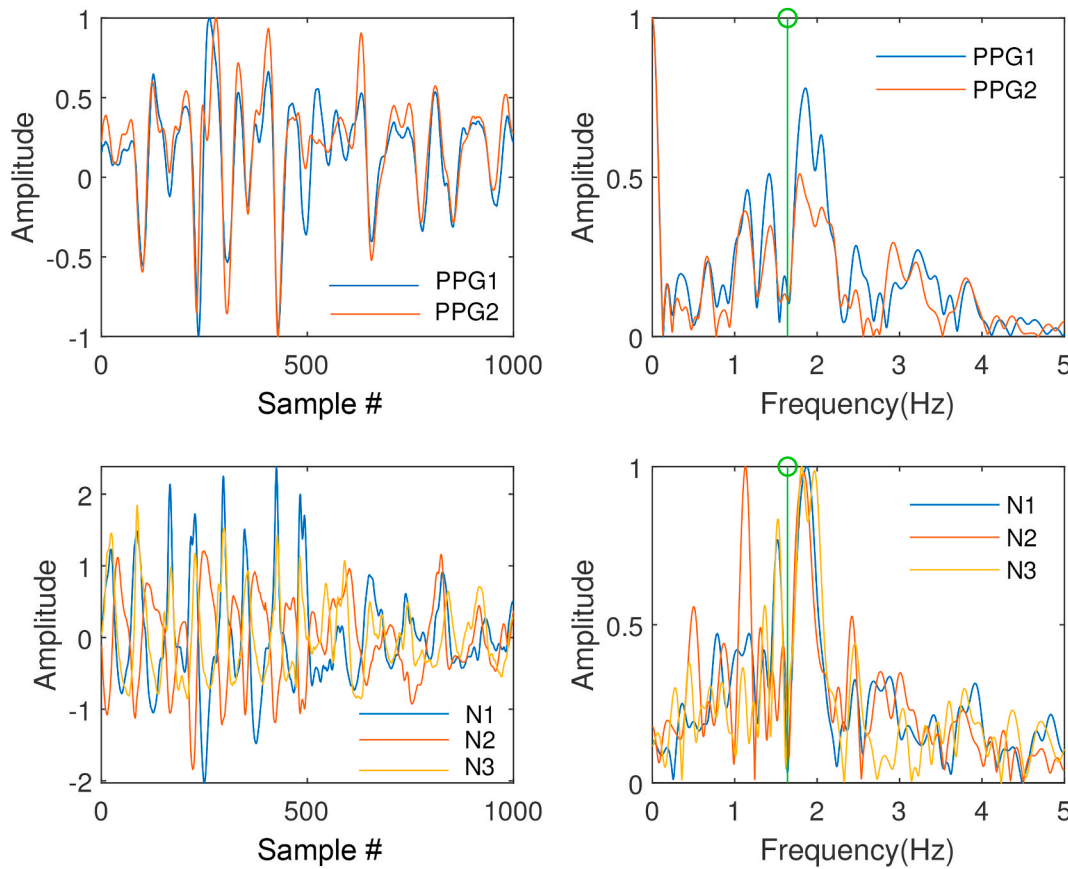


Fig. 2. Effect of MAs on PPG signals. The signal itself is composed of 2 x PPG and 3 x ACC channels, where PPG1 & PPG2 refer to PPG and N1–N3 are acceleration (ACC) signals of the three-axis accelerometer, the green line represents the actual heart rate. The effect of MAs on PPG channels is shown, none of the major spectral peaks in PPG1 or PPG2 refers to actual heart rate. Very small peaks in PPG1 and PPG2 are referring to actual heartbeats which are masked by much larger peaks due to MAs. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

spectrum analysis (SSA), temporal difference (TD), sparse signal reconstruction (SSR) using Bayesian learning and finally frequency bin selection. Based on this work, TROIKA [23], a complete framework for heart rate analysis was introduced. Key aspects of TROIKA included BPF, SSA, TD, focal under-determined system solver (FOCUSS) and spectral peak tracking. The study proved to be a major breakthrough and introduced the IEEE Signal Processing Cup 2015 (SPC) [24] dataset which, since then, has become a gold standard dataset for heart rate analysis. Later, the same dataset was employed by their group in a number of studies on PPG signal analysis [25–29]. The work in Ref. [25] is similar to TROIKA except that instead of using a single spectrum from acceleration signal, it uses all three acceleration channels resulting in a joint spectrum. Garde et al. [30] employed empirical mode decomposition (EMD) to decompose the signal before calculating power spectral density (PSD). A statistical analysis is then carried out to calculate heart and respiration rates. CurToSS [31] combined SSA with curve tracing but rather than using SPC, the authors employed PPG-DaLiA [32] and WESAD [33]. In Ref. [26], ensemble EMD (EEMD) was combined with spectral subtraction. The above discussion reveals that a major proportion of studies relying on multi-resolution analysis are based on EMD and its variants.

Signal conditioning is another major approach for heart rate estimation. Using this approach, authors in Ref. [34] introduced the SpaMA algorithm that constructs a clean PPG signal using only four spectral peaks (three peaks from the PPG spectra and one major peak from the acceleration spectrum). The reconstructed clean PPG signal is then employed for HR tracking. The algorithm was later extended to SpaMa plus by Reiss et al. [35]. The extension included tracking of heart rate using average of multiple windows in cases where the HR was uncertain.

Schack et al. [36] introduced a real time HR tracker that exploits the auto as well as cross correlation between the PPG signals. Likewise, Sun et al. [28] employed high resolution grid-less spectrum calculation and spectrum subtraction for heart rate estimation. Studies by Giorgio et al. [19,37] proposed a technique named as CARMA (Closest-subspace Algorithm for Reducing Motion Artifact), primarily targeting to reduce the computational complexity. The technique relies on LP and BP filtering, Hankel, SVD and Fast Fourier Transform. In a subsequent work, authors complemented CARMA with KNN [38] and introduced genetic algorithms for optimization of tracking parameters. Likewise, the authors in Ref. [27] employ asymmetrical least squares to suppress negative peaks resulting due to spectral subtraction of acceleration from PPG. Hardware-based signal conditioning is also employed to make the PPG signals fit for heart rate estimation. In one such study, Asada et al. [39] introduced the prototype of a wearable ring-based PPG sensor. Wearable finger sensors are one of the most widely employed bio-sensors in the field of PPG recording and were also used by Selvaraj et al. [6] to study the heart rate variability.

Adaptive filtering is another common technique that is investigated in a number of studies [29,40,41]. In Ref. [29], non-linear (Volterra) filtering was employed to remove the motion artifacts. Heart rate was then calculated and tracked using SSA and peak tracking. Motin et al. [40] employ recursive Wiener filtering for heart rate estimation. Experimental study is carried out on 23 signals from the IEEE SPC dataset and error rate of less than 2 bpm is reported. In another recent work, Chen et al. [41] first employ adaptive filters (LMS, RLS) to attenuate motion artifacts before using periodogram and peak selection for heart rate tracking.

A critical review of the literature discussed in the preceding

paragraphs reveals that a major proportion of techniques are based on signal decomposition like wavelet [21], EMD [23] and its variants [42]. While these methods have shown promising results, the computational load of such techniques is relatively high. To reduce the computational load, correlation based techniques [36] and techniques like SpaMa [34], CARMA [19] etc. are introduced which are basically signal conditioning methods. Studies like [21,23,26,27,30,43,44] etc. primarily focus on accurate heart rate estimation and are more suited for offline processing. On the other hand, techniques presented in studies like [19,28,36–38, 40] primarily target real time processing. Though signal processing-based methods have shown effective performance, a recent trend is to investigate machine learning-based methods for the heart rate estimation and similar problems as discussed in the following.

2.2. Machine learning-based techniques

The PPG signal analysis techniques discussed in the previous section are based on classical signal processing approaches. However, in the recent years, a paradigm shift has been observed towards supervised (machine) learning based methods. Such techniques rely on extraction (or learning) of features characterizing the signals. These features are then employed to train a learning algorithm (classifier or regressor) for a given task. The performance of such supervised methods depends on the availability of labeled training data. In case of biomedical signal analysis, though training samples are generally limited, machine learning based solutions have still gained significant popularity [20,35,45–52]. Notable classifiers employed for HR estimation include artificial neural networks (ANN), convolutional neural networks (CNN), long short-term memory (LSTM) networks, support vector machine (SVM) and random forest (RF) etc.

Among conventional machine learning techniques, support vector machine has remained a popular choice for PPG-based heart rate estimation (classification of the spectral peak associated with HR). Sun et al. [28] and Xiang et al. [53], for instance, trained an SVM using spectral peak separation and spectral peak ratio. Likewise, Ye et al. [50] proposed a random forest based approach for spectral peak tracking to locate the spectral peak corresponding to HR. In another study, Zhu et al. [51] employed spectral subtraction with a neural network for HR tracking in the presence of MAs and noise. Roy et al. [52] introduced a personalized system that exploited a single PPG beat rather than the complete signal. During training, two multi-layer perceptrons were trained. The first of these assessed the quality of signal (clean or noisy) and also generated a reference beat template while the second network employed a set of features produced by an auto-encoder from PCA-reconstructed data.

Like many other pattern classification tasks, in the recent years, a major shift in supervised learning-based techniques from conventional machine learning to deep learning can be observed for the heart rate estimation problem as well. Deep neural networks like de-noising convolutional neural networks (DCNNs), temporal convolutional networks (TCNs), neural architecture search (NAS) etc. have been commonly investigated for different stages of processing. Among notable deep learning-based techniques, an interesting work is reported in Ref. [49] which employs EEMD and LSTMs for reference heart rate generation while VMD is used for heart rate tracking. Reiss et al. [35] exploit CNN while Biswas et al. [20] investigate the combination of CNN and LSTM for heart rate estimation. Unlike Reiss et al. [35] where the PPG signal is first converted to an image using STFT, Biswas et al. [20] as well as related studies like PPGnet [47] and PPnet [48], directly feed the signal to convolutional layers for feature extraction. An LSTM network is then employed to estimate the heart rate from the learned features. While PPGnet only estimates the heart rate, PPnet also provides the diastolic and the systolic blood pressure (DBP and SBP). A number of recent studies [45,46] targeted the problem of reducing the computational load in machine learning-based solutions. Instead of employing real-numbered values for weights (and biases), usage of binary weight and bias values is

proposed. The resulting system is named as Binary Cornet (b-Cornet) which employs a binary CNN (b-CNN) and a binary LSTM (b-LSTM).

Among other studies, DeepHeart was proposed in Ref. [54] and employs DCNNs for de-noising the PPG channels. Risso et al. [32] and Burrello et al. [55] employed TCNs but it should be mentioned that TCNs are basically off-the-shelf hardware modules. Song et al. [56] employed Neural Network Intelligence (NNI) before feeding signals to deep network. In all of these studies, authors have applied heavy pre-processing before feeding the signal for classification.

From the discussion above, it can be seen that the preliminary attempts to machine learning-based heart rate estimation primarily employed classical learning methods like SVM, RF, ANN etc. [28, 50–53]. Subsequently, deep networks like CNN, LSTM etc. and their combinations [20,35,45–49] attracted significant research attention. In the recent years, off-the-shelf hardware modules like DCNNs, TCNs and NNIs are being used for these networks and major focus of recent studies is the real time applications of heart rate estimation [32,54–56].

An overall analysis of techniques reported for heart rate estimation from PPG signals reveals that the recent literature is dominated by machine learning-based methods. These supervised techniques rely on availability of labeled training data and, making the algorithm learn during training phase is known to be a resource-hungry phase. However, the model, once trained and transferred to platforms like Raspberry Pi etc., is able to provide inference in real time. Table 1 presents an overview of well-known (signal processing and machine learning-based) heart rate estimators. Heart rate estimation is a complex process which is composed of multiple stages of pre-processing, feature extraction and learning from raw signals or features. Moreover, in order to estimate heart rates with low error, we need additional measures like restriction of solution space and use of heuristics etc. The quantification measures generally used are Error1 and Error2 which are based on the mean absolute error (between predicted and ground truth heart rate) and are formally defined later in the paper.

The comparative table also shows that the errors reported by signal processing and machine learning based methods are comparable in many cases. Signal processing techniques mostly require pre-processing steps limiting the solution space to spectral range between [0.4–6] Hz. These estimators generally use exercise model and ACC signals to de-correlate the heart rate from MAs. Furthermore, these estimators are commonly generalized and not subject-specific. Machine learning-based methods, on the other hand, are mostly subject-specific but they do not limit the search space to the general heart rate spectra. Such methods, in general, tend to be more robust and are likely to estimate multiple biometric markers simultaneously in the future [20]. A comprehensive review of heart rate estimation from PPG signals can also be found in our previous work [58].

3. Dataset

The experimental study of our system is carried out on the benchmark IEEE Signal Processing Cup (SPC) dataset [24]. IEE SPC is a publicly available PPG dataset that has been widely employed for algorithmic development and evaluation. The dataset is composed of three parts and the first set of 12 samples was made available in August 2014. Another set of training samples as well as the test set of 10 samples was provided in January 2015. The training samples have 6 channels where the first channel corresponds to ECG reference, the next two channels are PPG and the last three refer to the acceleration (ACC) signals (x,y and z axis). The ECG reference, however, is not provided in the test samples. PPG signals have been acquired using pulse oximeters while the acceleration signal was recorded from a three-axis accelerometer. The ECG reference was simultaneously acquired from the chest using ECG sensors. All signals (ECG, PPG and ACC) are sampled at a sampling frequency of 125 Hz. Two types of exercises are used and the sequence within these exercises is outlined in Table 2. Type-01 exercises which constitute the 12 signals in the training set are primarily whole

Table 1

An overview of prominent HR estimation techniques ('SP' refers to signal processing while 'ML' refers to machine learning based solution).

Estimator	Estimation Method	Classifier	HR Stages	Remarks	Error1	Error2
Chen et al. [41], 2021	SP	–	LMS & Adaptive filtering (LMS, RLS), Periodogram and peak selection	Multichannel PPG signals & Heart rate estimation by scoring method	1.16 & 1.56	0.87
CurToSS [31], 2020	SP	–	Sparse Spectrum Analysis & Curve tracing	Time-frequency analysis of PPG and ACC.	1.93 & 3.56	2.07
Schack et al. [36], 2017	SP	–	Maximization of cost function	Down sampling of PPG and ACC signals, cross correlation and Gaussian filtering, cost function based tracking	1.32	–
Emroz et al. [42], 2016	SP	–	Comparison of heart rate from multiple sources	Building of HR space, comparison of two sources for HRC at a time & fine tuning	1.02	–
TROIKA [23], 2015 & JOSS [25], 2015	SP	–	Initialization, Peak selection, Peak Verification & Smoother algo. (JOSS)	Search areas for F0 & F1 in PPG spectra	2.42 & 1.28	1.82 & 1.01
CC [26], 2015	SP	–	Heuristics	Two spectral peaks in PPG; Considers direction and direction of error	1.83	1.40
DeepHeart [54], 2021	ML	DCNN & Ensemble	Deep-learning-based denoising and spectrum-analysis-based calibration	Simultaneous use of ECG and PPG signals	1.61	–
Risso et al. [57], 2021	ML	TCN & NAS	Use of MorphNet, TimePPG & Post processing	Use of TEMPONet & PPGDalia signals	3.84	–
Q-PPG [55], 2021	ML	TCN	Use of MorphNet, PIT and EdMIPS	Use of Seed Network & PPGDalia signals	4.41	–
NAS-PPG [56], 2021	ML	DNN	NNI & NAS-PPG model	Use of SPC & PPGDalia signals	6.02	–
Rocha et al. [45], 2020	ML	DNN	Two layers of Binary CNN and Binary LSTM	FPGA implementation of Binary Cornet	3.75	–
Zhu et al. [51], 2019	ML	NN	Neural network modelling, smoothing and linear regression	Polynomial approximation of false heart rate	1.03	0.79
Biswas et al. [20], 2019	ML	CNN	CNN extracted features with LSTM	Two layers of CNN and LSTM	1.96	1.47
Roy et al. [52], 2018	ML	MLP	Trained NN using features generated by auto-encoder from PCA	Generate HR template using clean PPG	1.47	1.10
SpaMa [34], 2015 & SpaMa Plus [35], 2018	ML	CNN	Heuristics & trained a CNN using 4 channel images converted from signals	Down sampling of PPG and ACC. Three spectral peaks in PPG & ACC and iterative selection of spectral peak; End-to-end learning	1.93 & 3.56	2.07
Sun et al. [28], 2016	ML	SVM	SVM trained on peak amplitude and peak-to-peak separation	–	1.78	1.57

Table 2

Exercise types and their sequence in IEEE Signal Processing Cup (SPC) 3.

	30s	1 min	1 min	1 min	1 min	30s
TYPE-01	Rest	8 km/h	15 km/h	8 km/h	15 km/h	Rest
TYPE-02	Rest	6 km/h	12 km/h	6 km/h	12 km/h	Rest

body exercises while Type-02 exercises (10 signals in the test set) are mainly the upper body exercises involving fists and arms. Both types of exercises start and end at rest and there is a total of 4 min of exercise in between the resting conditions.

In the present study, both subject-dependent and subject-independent models are employed for heart rate estimation. For subject-specific models, a part of the signal of a subject is used for training and the remaining samples are utilized in the evaluation phase. Hence, all the signals in the IEEE SPC dataset are used during training and test phases. On the other hand, for subject-independent study, a number of different experimental protocols are employed including different combinations of Type-01 and Type-02 exercises. More details on the distribution of data in subject-independent experiments can be found in Section 5.2.

4. Methods

This section presents the details of the proposed heart rate estimation technique that is modeled as a supervised learning problem. As discussed earlier, signal processing-based techniques are generally accompanied with a number of heuristics and (empirically determined) parameters. Although these heuristics as well as the chosen parameter values can be intuitively justified, it may, to some extent, prevent the

generalization of the proposed methods. Consequently, we have modeled the HR estimation as a *regression* problem. We propose a machine learning-based hybrid solution that relies on a combination of hand-engineered and data-driven feature extraction. More specifically, a combination of convolutional and recurrent neural networks (C-RNN) is employed, the convolutional part learns robust feature representations while the recurrent part treats the extracted features as a sequence modeling problem and learns the mapping from the input sequence to the heart rate. An overview of key processing steps in HR estimation using the proposed machine learning-based technique is presented in Fig. 3.

The signal is first filtered and is divided into windows in the pre-processing steps. Each window can be fed directly as raw data to the convolutional layers to learn features from raw values which can be subsequently fed to the recurrent layers for sequence modeling. We, however, first convert the raw data into features and then feed these feature sequences to the 1D convolutional layers to learn high level robust feature representations. This represents a hybrid approach where representation learning is carried out on features rather than raw signal values and allows exploiting the advantages of both signal processing and machine learning. The learned representations are then employed to train the recurrent layers. Once the model is trained, it is fed with test samples for heart rate estimation. Details of each of these processing modules are presented in the following sections.

4.1. Pre-processing

Prior to feature extraction and subsequent analysis, the PPG as well as ACC signals are pre-processed with a band pass filter. The objective of pre-filtering is to limit the low and high frequency noisy components

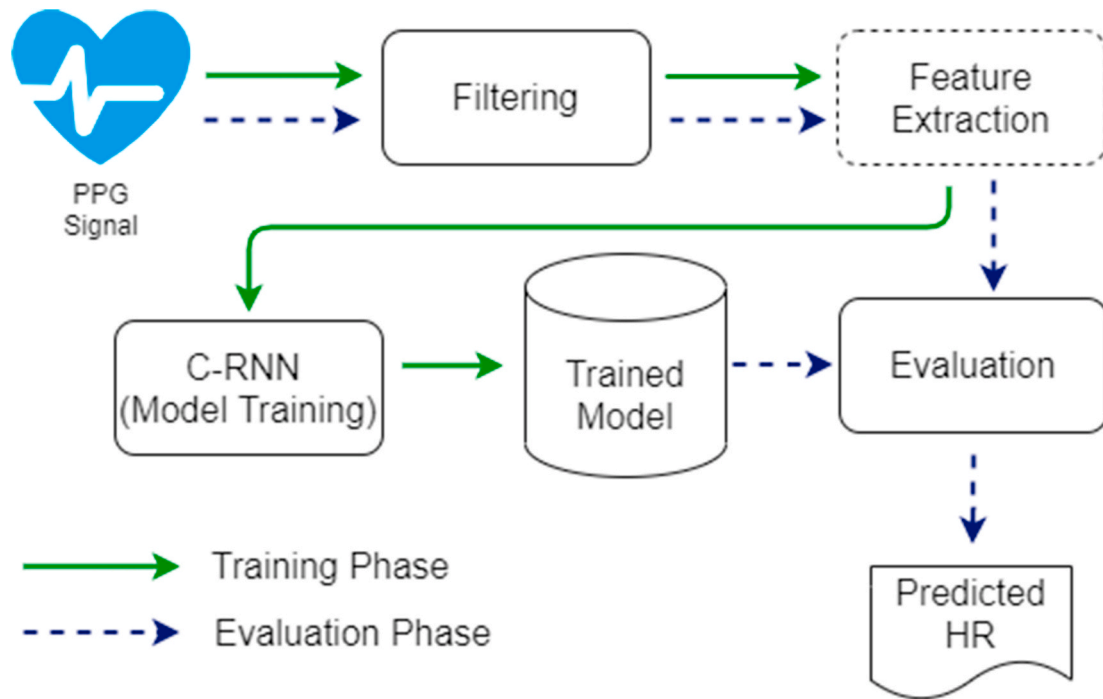


Fig. 3. Overview of key processing steps for heart rate estimation. Estimation is divided into training and test phase. Once the regressor (C-RNN) is trained using features, it estimates heart rates during inference.

which do not reflect heart rate as a heart rate of below 30 bpm or above 300 bpm is generally not possible due to the bio-mechanical nature of human heart. Hence, employment of a bandpass filter with spectral response in the range between 0.4 and 5 Hz is likely to estimate the heart rate with high accuracy. However, we investigated a number of filters in our study as listed in Table 3 and the magnitude response of different filters is illustrated in Fig. 4. The filters f_1 and f_2 are selected based on studies like Troika [22,23,26] and the bandwidth of these filters reflects the general range of human heart rates. Filter f_3 also accommodates the harmonics which may appear in addition to the fundamental frequency related with heart rate. Finally, filters f_4 and f_5 are inspired from the work in Ref. [20] which employs neural networks similar to the present study.

The impact of the choice of filter on the overall performance (training models and subsequently estimating heart rate) is also illustrated in Fig. 4. It can be seen that the performance of filters varies when applied to different signals. For example, the error reported by filter f_2 is minimum for all the signals except Signals 4–6. Similarly, filter f_5 has reported very low error for the Signals 4–6 and the Signals 11–22. Application of filters f_3 and f_4 , in general, results in relatively high errors. Overall, filter f_1 outperforms other filters and reports the least error

Table 3

Summary of different filters investigated to pre-process the PPG signals. Table lists the filter types as well as their order and spectral range. The reason for choice of specific type of filter is also mentioned.

Filter	Filter Type	Order	Bandwidth	Remarks
f_1	Elliptical	30	[0.4–5]	General range used for heart rate estimation
f_2	Butterworth	2	[0.4–5]	General range used for heart rate estimation
f_3	Elliptical	30	[0.4–10]	Heart rate range is extended to account for harmonics of fundamental frequency
f_4	Butterworth	4	[0.1–18]	Filter employed in [20]
f_5	Elliptical	30	[0.1–18]	Higher order Elliptical implementation of filter introduced in [20]

in most cases. Quantitatively, errors are presented in terms of average absolute error and percentage absolute error in Fig. 4-c and Fig. 4-d respectively. Average error reported by filter f_1 is in the range 2–5 bpm. On the other hand, the error reported by other filters varies between 1 and 20 bpm. The variation in the error reported by f_1 is also low as compared to the other filters reflecting the stability of this filter. The least error reported by f_1 also validates the argument that a filter which has bandpass characteristics in the range between 0.4 and 5 Hz is able to estimate heart rates with high accuracy.

4.2. Feature extraction

Features are extracted from two PPG and three acceleration channels using windows of $N_{smp} = 1000$ which, with a sampling rate of 125 Hz, corresponds to 8 s. It is important to mention that we employ both the PPG channels which increases the probability of finding clean patches among the two channels. Since the path to photo-detector from light emitting diode is different for different PPG channels, hence, motion artifacts differently affect the two PPG channels and the likelihood of finding clean patches becomes high. For all features except the spectral features, the 1000 raw signal values are first re-shaped (row-wise) into a matrix of $n_{feat_sub} \times N_{feat}(8 \times 125)$ resulting in feature vectors of dimension N_{feat} as shown in Fig. 5. The spectral features are computed using the N_{smp} dimensional vector and also mapped to a vector of length N_{feat} like other features (Fig. 5). Features are computed in the fractal, statistical, time, energy, EMD-based modes and frequency domains and are elaborated in the following. A summary of parameters employed in computation of these features is presented in Table 4 while the details of these features are introduced in the following.

4.2.1. Fractional Brownian motion feature

To calculate the fractional Brownian motion (fBm), the signal is assumed to be a realization of continuous time Gaussian process. To quantify fBm, three Hurst (H) parameters are used. H value of 0.5 is Brownian motion and its derivative is the white noise. $H > 0.5$ shows the association of long term memory with the signal while $H < 0.5$ is for signals with short term memory. The first two Hurst parameters are

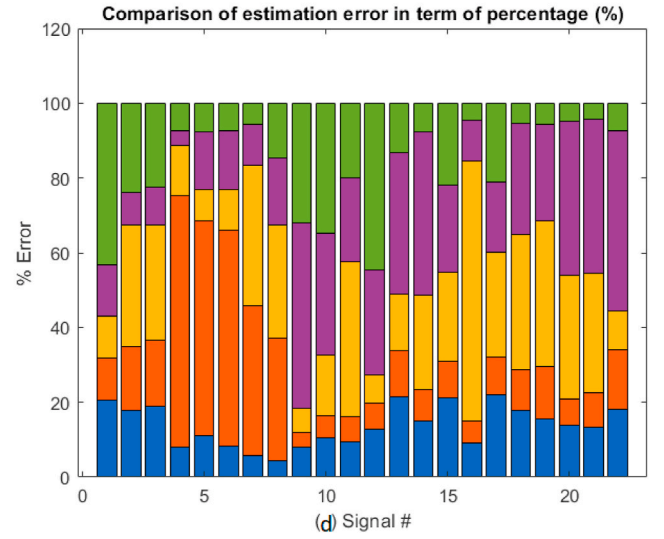
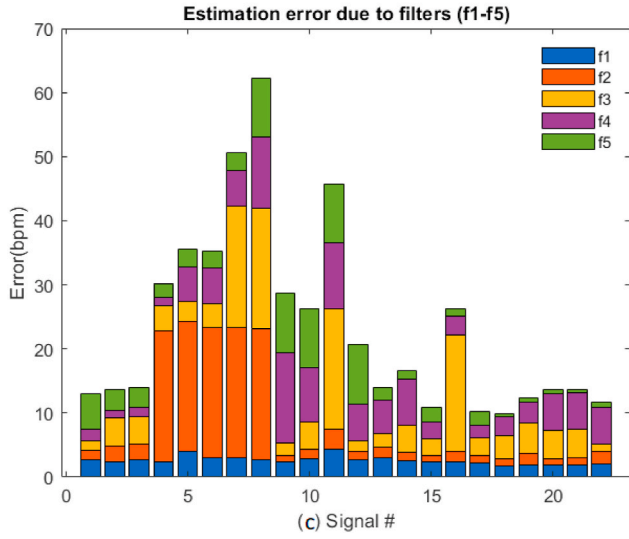
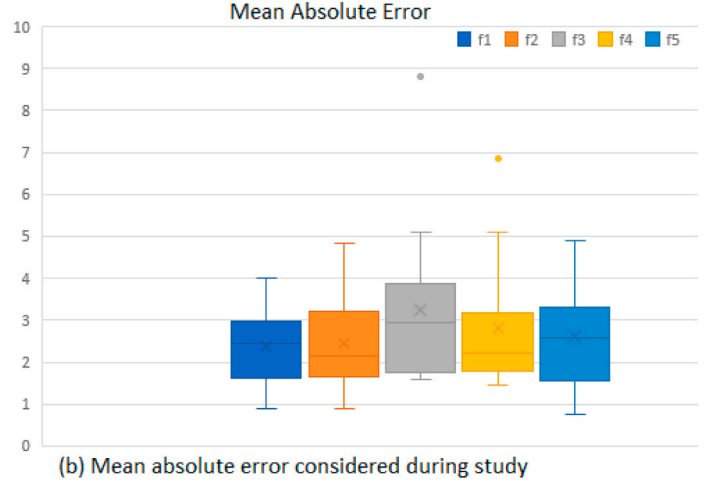
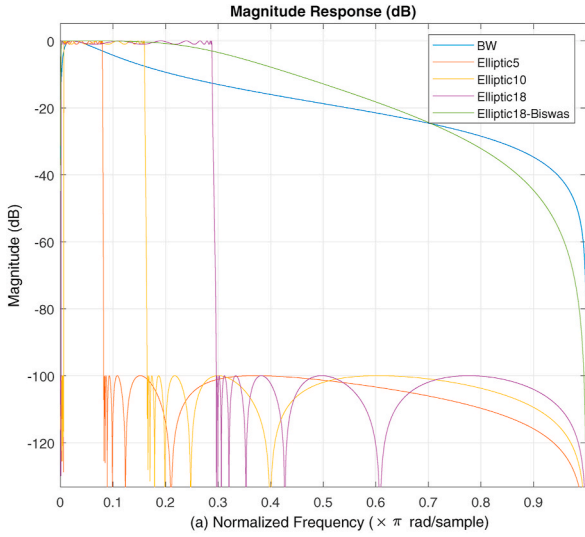


Fig. 4. Magnitude response and the resultant estimation error of the filters (f1-f5). (a) Magnitude response of the filters in decibels (dB). (b) Mean absolute error of different filters (f1-f5) considered in our study. (c) Estimation error in terms of bpm. (d) A comparison of errors of different filters.

calculated using the discrete second derivative while the third parameter is based on linear regression of variance. In our study, we have employed the first and the third parameter as features.

4.2.2. Statistical features

The statistical features considered in our study include the coefficient of variation (ratio of standard deviation to mean), mean, and the range of signal. It must be noted that the global mean is zero after bandpass filtering with f1. However, the local mean and subsequently the coefficient of variation are not zero as they are calculated using a small window of 8 s. Furthermore, Hjorth parameters are also employed as features. Hjorth parameters refer to activity, mobility and complexity. Activity represents the variance of signal while mobility and complexity are defined in the following.

$$\text{Mobility} = \sqrt{\frac{\text{var}(y'(t))}{\text{var}(y(t))}} \quad (1)$$

$$\text{Complexity} = \frac{\text{Mobility}(y'(t))}{\text{Mobility}(y(t))} \quad (2)$$

Where $y'(t)$ is first order derivative of signal with respect to time. Mobility and complexity are employed to capture information on the

mean frequency of signal and, signal similarity to a pure sine wave.

4.2.3. Time domain features

Time domain features considered in our work include the signal sub-sampling, Shannon energy and EMD decomposition.

- **Signal sub-sampling:** Signal is converted from N_{smp} samples to $N_{\text{feat}} \times n_{\text{feat_sub}}$ matrix and column-wise summation is carried out to convert the matrix in to an N_{feat} dimensional vector.
- **Shannon energy features:** Shannon entropy and Shannon log energy are calculated from the signal and are converted to vectors of length N_{feat} .
- **EMD-based features:** The signal is first decomposed in to $L_{\text{EMD_dec}}$ levels using EMD decomposition and then, $L_{\text{EMD_ref}}$ levels are used as features. Low and high frequency channels represent the frequencies other than heart rate, hence are discarded.

4.2.4. Spectral features

For spectral features, we first calculate the Chirp Z transform (CZT-based) PSD of length N_{PSD} and subsequently compute three features namely full power correlation feature, power feature and partial power correlation feature.

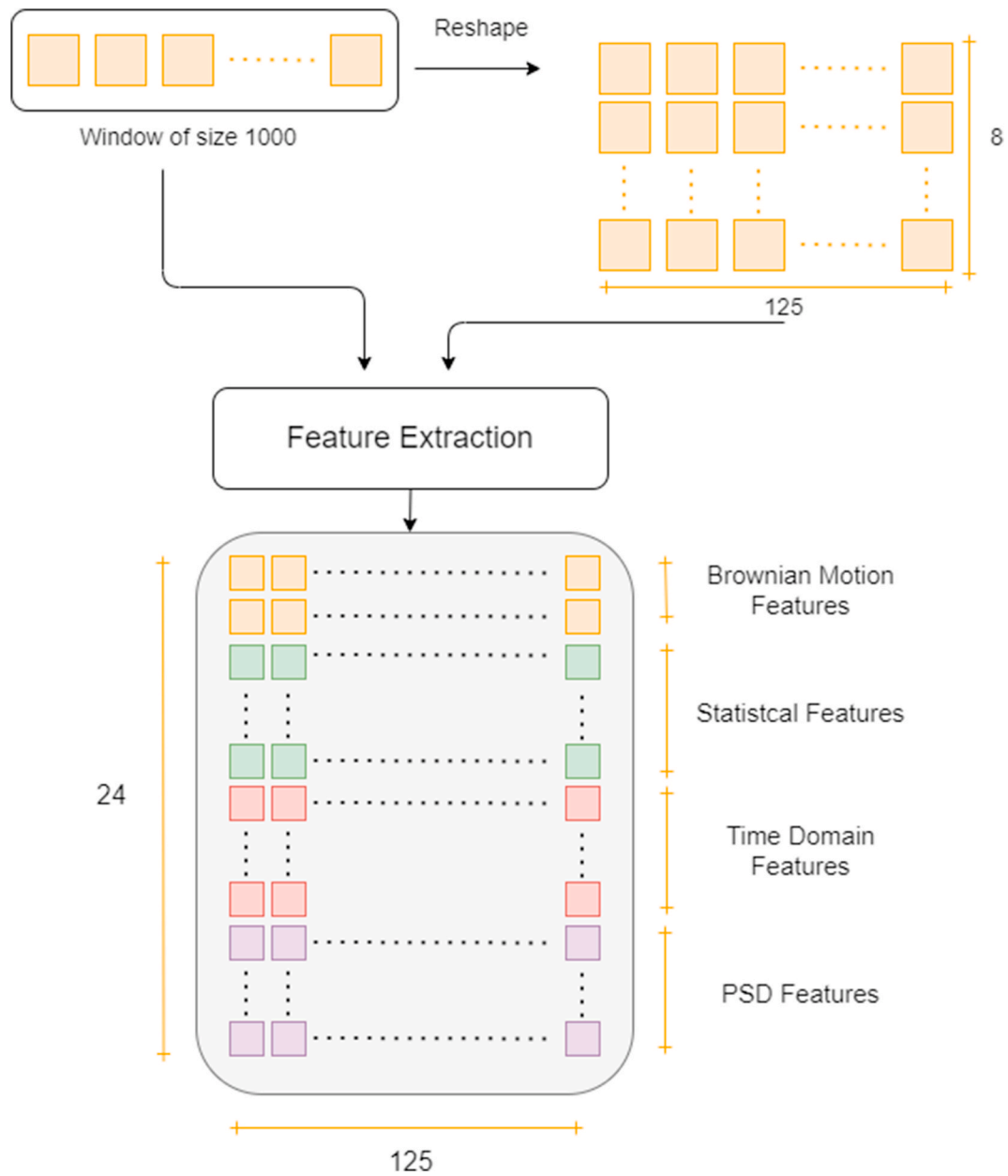


Fig. 5. Reshaping on input windows for feature extraction. EMD and PSD Features are computed from the window of 1×1000 while other features are calculated from the re-shaped matrix of size 8×125 . Feature extraction produces a matrix of 24×125 . Applying the same feature extraction to 2 PPG and 3 ACC channels produces a final data of 120×125 .

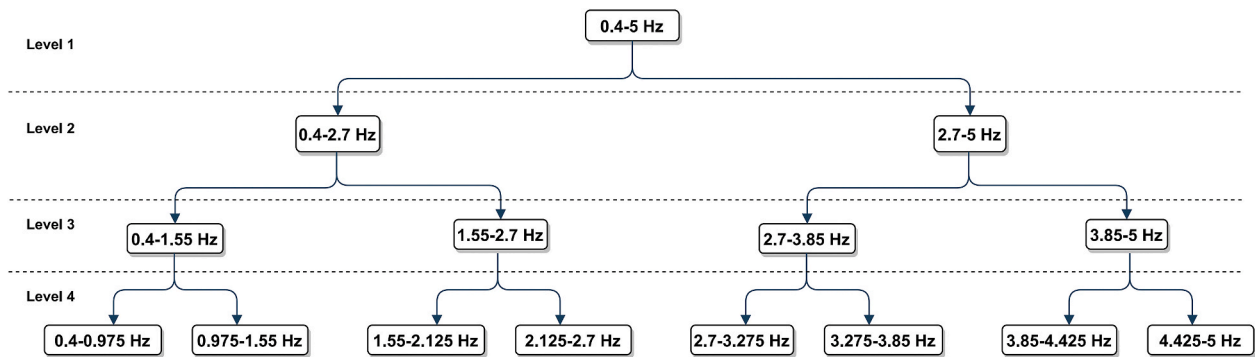


Fig. 6. Dyadic representation of PSD.

Table 4

Summary of parameters employed for feature extraction.

Parameter	Value	Remarks
N_{smp}	1000	General length of signal ($2 \times \text{PPG}$, $3 \times \text{ACC}$)
N_{feat}	125	Length of feature vector
n_{feat_sub}	8	Length of feature sub-window
L_{EMD_dec}	12	Total decomposition levels of EMD
L_{EMD_ref}	[5–8]	Mid-band frequency channels of EMD

- **Full power correlation feature:** The unbiased auto-correlation of length $2 \times N_{feat} - 1$ is calculated which is converted into Toeplitz matrix. Pearson's correlation is then calculated from Toeplitz matrix and finally, Eigen decomposition is performed which gives Eigen values of length N_{feat} , employed as features.
- **Power feature:** The PSD is represented at L_{sp} levels and the relationship between subsequent levels is dyadic in nature as shown in Fig. 6. Power feature is calculated for $L_{sp} > = L_{sp,ref}$ for further processing. For $L_{sp} = L_{sp,ref}$, power vector remains the same. However, for $L_{sp} > L_{sp,ref}$, power samples are added up in such a way that the total length of power feature becomes equal to N_{feat} .
- **Partial power correlation features:** Calculation of partial correlation feature is same as full correlation feature but this feature is calculated at $L_{sp} > = L_{sp,ref}$. However, since, the length at each level of $L_{sp} > = L_{sp,ref}$ is different, hence after Eigen decomposition, Eigen values are added up to give a vector of length N_{feat} .

A summary of the features employed in our study is presented in Table 5. The choice of spectral features is motivated by their effectiveness in estimating heart rate as validated in a number of studies [23,25,41,42,59]. Likewise, statistical and time-based features are employed to capture the temporal properties in the signal. Finally, it is known that heart rate estimation is a memory-based process as heart rate cannot exhibit a sudden change due to bio-mechanical constraints, hence, fractional Brownian motion features are used to model this property.

4.3. Sequence modeling: C-RNN

The recent advancements in deep learning have resulted in a paradigm shift from conventional classification and regression frameworks to a combined feature extractor (selector) and classifier/regressor. A key milestone that led to the current popularity of such methods was the investigation of convolutional neural networks for image recognition task on the ImageNet challenge [60]. Since then, CNNs are considered gold standard for automatic feature learning from image, text and signal data. Likewise, for analysis of sequential (time series) data, recurrent neural networks have been widely investigated [61]. Contrary to the conventional feed forward networks, RNNs employ recurrent units where the activation of current (hidden) state not only depends on the current input but also on the activation(s) from previous states as summarized in Equation (3).

$$a_i^t = g(W_x \cdot x_i^t + W_a \cdot a_i^{t-1} + b_a), \quad (3)$$

Table 5

Summary of Features employed in our study.

Feature	Description	Dimensionality
Fractional Brownian Statistical	Hurst Parameters	2×125
	Statistical Measures	3×125
	Hjorth Parameters	3×125
Time Domain	Sub-Sampling	1×125
	Shannon Energy	2×125
	EMD	4×125
Spectral Features	Power & Power Correlation	9×125
Total-Single Channel	Combined features for one channel	24×125
Total-All 5 Channels	$2 \times \text{PPG} \& 3 \times \text{ACC}$ channels	120×125

The activation a_i^t at a given time step t is functionally dependent on the input x_i^t as well as the activation of previous time step a_i^{t-1} . W_x and W_a represent the learnable parameter matrices while b_a is the bias vector. The output is calculated using the activation(s) of the hidden state(s) as follows.

$$y_t = g(W_y \cdot a_t + b_y) \quad (4)$$

For many-to-many mappings the output y_t is computed at multiple time-steps while for many-to-one problems (as in our case), the output is calculated at the final time-step only. The functional dependencies of hidden layers on previous hidden layers (in depth as well as time) allow RNNs to exploit the contextual dependencies in the input. Though RNNs effectively capture the temporal patterns in time series data, the vanishing (exploding) gradient problem prevents them from learning long-term dependencies in the data. Consequently, it is common to employ different variants of simple RNNs, Long-Short Term Memory (LSTM) networks being the most popular among these.

LSTMs introduced in Ref. [62] are able to learn long-term dependencies in the input. A typical LSTM cell (Fig. 7) comprises of *gates* which can add or remove information from the LSTM cell state. Gates are implemented through a sigmoid layer and point-wise multiplication. The output of the sigmoid layer ([0 1]) regulates the flow of information in the cell state. The LSTM cell has three such gates, the forget gate, the input gate and the output gate. The forget gate controls the information to be removed from the cell state while the input gate decides which values in the cell state should be updated (as a function of previous hidden state h_{t-1} and current input x_t):

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (5)$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (6)$$

To update the cell state, candidate values are produced by a *tanh* layer and the cell state is updated.

$$\tilde{C}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c) \quad (7)$$

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \quad (8)$$

Output of the cell h_t is regulated by the output gate:

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (9)$$

$$h_t = o_t \odot \tanh(C_t) \quad (10)$$

In our study, we employ a combination of convolutional and recurrent (LSTM) layers to exploit the time series information in the signal (feature sequences) which can be employed to estimate the heart rate. The general structure of the proposed model is outlined in Fig. 8 while the architectural details are summarized in Table 6. Details of different layers in our model are discussed in the following.

- **Conv Layers:** The feature sequences are fed to a stack of two one-dimensional convolutional layers. Each layer employs a convolution kernel of size $f = 3$ with a stride of $s = 2$. These layers not only serve to extract local temporal dependencies in the input but also down sample the input (stride > 1). The first conv layer employs 32 while the second layer uses 64 convolutional filters.
- **Batch Normalization Layers:** Batch Normalization (BN) normalizes the activations of a layer using batch mean and batch variance along with two learnable parameters. The primary objective of batch normalization is to speed up training by limiting the internal co-variate shift.
- **Dropout Layers:** Dropout is a well-known regularization technique that is aimed at preventing over-fitting. A dropout layer drops each neuron in the network with a probability p during training. This ensures that the model performance is not too much dependent on a specific set of neurons which in turn serves to control over-fitting. In

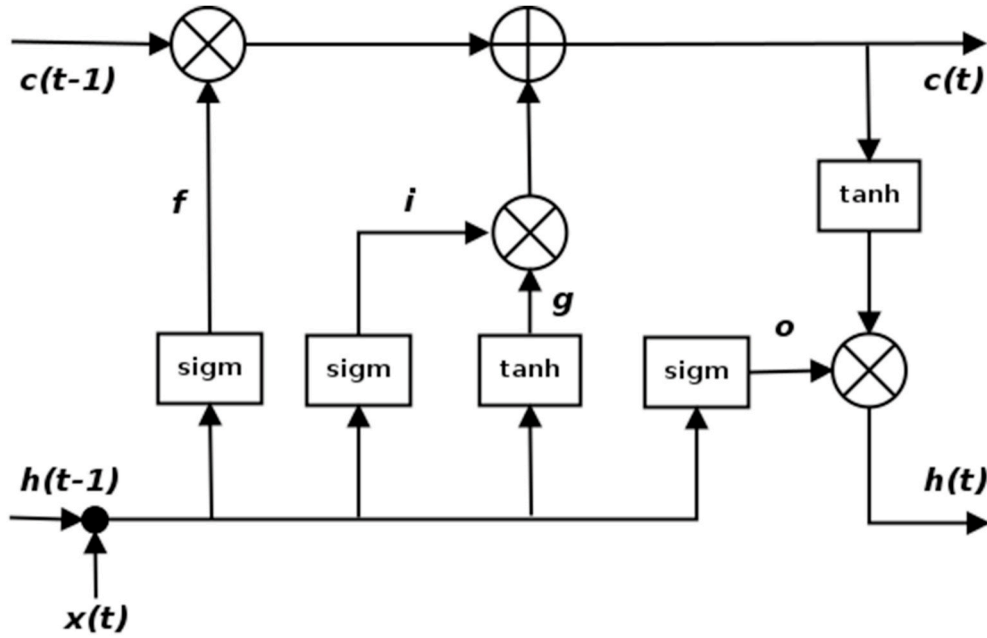


Fig. 7. Structure of a typical LSTM cell.



Fig. 8. C-RNN model employed for HR estimation.

Table 6
Architectural details of the C-RNN model employed in our study.

Layer	Hyper Parameters	Output Size	Parameters
Conv_1	Filter size = 3, stride = 2, filters = 32	60×32	12,032
BN_1	–	60×32	128
Dropout_1	Dropout rate = 0.2	60×32	0
Conv_2	Filter size = 3, stride = 2, filters = 64	30×64	6,208
BN_2	–	30×64	256
Dropout_2	Dropout rate = 0.2	30×64	0
LSTM_1	LSTM units = 128	30×128	98,816
Dropout_3	Dropout rate = 0.2	30×128	0
LSTM_2	LSTM units = 64	30×64	49,408
Dropout_4	Dropout rate = 0.2	30×64	0
LSTM_3	LSTM units = 32	32	12,416
Dense_1	Units = 1	1	33
		Total	179,297

our implementation, we introduce dropout layers with a dropout rate of 0.2 i.e. 20%.

- **LSTM Layers:** A cascade of three LSTM layers is employed in our model with 128, 64 and 32 units respectively.
- **Dense Layer:** The problem of heart rate estimation is posed in the framework of a many-to-one mapping. The last LSTM layer is (fully) connected to a single neuron which outputs the predicted heart rate.

- **Activation Functions:** The conv layers employ *ReLU* while the LSTM layers employ *tanh* as the activation function. The final dense layer implements the regressor with a linear activation function.

The model is trained in an end-to-end fashion where a batch of windows is fed to the network along with the ground truth heart rate. Mean absolute error is used as the loss function and the network is trained using mini-batch gradient descent with a learning rate of 0.001. Evolution of loss on training and validation sets as a function of number of epochs for one of the subjects is illustrated in Fig. 9.

5. Experiments and results

A comprehensive series of experiments is carried out on the set of 22 PPG signals in the IEEE SPC dataset which is divided into training and test datasets comprising 12 and 10 signals, respectively. We carried out both subject-specific and generalized (subject-independent) experiments. For subject-specific experimental protocol, for a given subject, a subset of samples is employed for training while the rest of the samples of the same subject are used for inference. For subject-independent mode, a series of four different experiments is carried out. In the first two experiments, training is carried out on the 12 signals in the training set whereas the 10 signals in the test set are used for inference. Training and test sets reflect two types of exercises and the sequence within these exercise are elaborated in Section 3 and summarized in Table 2. Among the two experiments, the first one employs raw signals while the second

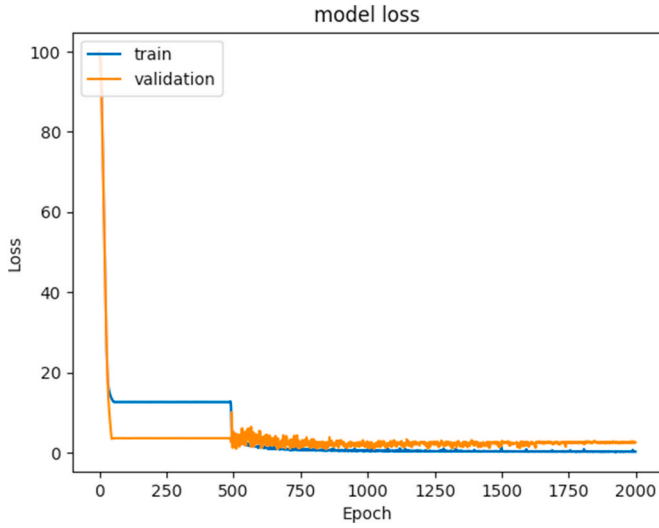


Fig. 9. Evolution of loss on training and validation data as a function of training epochs.

relies on features. In the third experiment, a subset of samples from the 10 signals are combined with the 12 signals in the training set to train the model while the remaining samples are used as test set. Likewise, the last experiment involves combining all the 22 signals in a single set and then randomly distributing the samples into training and test sets. A summary of the experimental setup for subject-independent evaluations is presented in Table 7.

We present the experimental results using each of these protocols in the following.

5.1. Subject-dependent protocol

Experiments are first carried out in a subject-dependent mode where 70% of the windows of each subject are employed to train the model, 10% as the validation and 20% as the test set. The evaluation metrics employed include the Mean Absolute Error (MAE or Error1) and Mean Absolute Percentage Difference (Error2) as given in the following equations.

$$Error1 = \frac{1}{W} \sum_{i=1}^W |BPM_{est}(i) - BPM_{true}(i)| \quad (11)$$

$$Error2 = \frac{1}{W} \sum_{i=1}^W \frac{|BPM_{est}(i) - BPM_{true}(i)|}{BPM_{true}(i)} \quad (12)$$

Where $BPM_{est}(i)$ and $BPM_{true}(i)$ represent the estimated and the ground truth value in window i and W is total number of windows.

Among the different filters investigated to pre-process the data, filter $f1$, i.e., 30th order Elliptical filter with bandwidth of [0.4–5] reported

Table 7

Experimental setup for subject-independent protocol. T_1 and T_2 refer to Type-01 and Type-02 exercises respectively, $Raw(T_1, T_2)$ and $F(T_1, T_2)$ represent experiments with Type-01 signals in the training and Type-02 signals in the test set using raw signals and features respectively. Both $F(T_1 + T'_2, T_2 - T'_2)$ and $F(M)$ refer to the experiments which employ features. $F(T_1 + T'_2, T_2 - T'_2)$ represents the scenario where a sub-set of test samples is combined with the training set while $F(M)$ refers to mixing the two types of exercises and randomly distributing signals into train and test sets.

Experimental Protocol	Training Samples	Test Samples	Signal/Features
$Raw(T_1, T_2)$	1768	1328	Raw Signals
$F(T_1, T_2)$	1768	1328	Features
$F(T_1 + T'_2, T_2 - T'_2)$	2418	678	Features
$F(M)$	1925	1171	Features

the lowest errors (as well as lowest variance in errors) as shown in Fig. 4-b. Consequently, subsequent experiments are carried out using the same filter to pre-process the signals.

We first discuss the performance of the C-LSTM model using raw data from two PPG and three acceleration channels i.e. we directly feed raw values to the model and let the two convolutional layers learn features directly from the data. Each window fed to the model comprises of length $N_{smp} = 1000$ i.e. 1000 time-steps and each time-step is represented by a vector of 5 values ($2 \times PPG + 3 \times ACC$) as illustrated in Fig. 10. It can be seen from Table 8 that average error values of 5.42 ± 6.27 and 5.06 ± 5.85 are reported on the 22 signals in the dataset for metrics mean absolute error (Error1) and mean absolute percentage difference (Error 2) respectively. While most of the signals report stable performance, few of the signals exhibit relatively high variation in the errors.

In the second experiment, rather than feeding raw data to the model, we feed the feature sequences to the model. Each window of 5×1000 of raw data is mapped to a window of 125×120 as elaborated in Section 4.2. The convolutional layers in this case serve to learn high level robust representations of already computed features. These layers also serve as feature selectors as convolution (which indeed is a dot product of filter weights with the input) can adapt the filter weights corresponding to different features as a function of their effectiveness. It can be seen from Table 8 that a significant performance enhancement is observed as opposed to raw features. Overall average errors of 2.41 ± 2.90 and 2.12 ± 3.15 are reported on the 22 signals in the dataset for evaluation metrics mean absolute error and mean absolute percentage difference respectively. It is interesting to note that not only the average error values have reduced as opposed to feeding raw signals, the performance of the features is relatively more stable with lesser variation in the errors. It is also important to note that the reduction of error when using features depends on the signal itself, as signals are differently affected by motion artifacts. Depending upon how corrupt a signal is with noise and artifacts, reduction in error varies from one signal to another.

In the next series of experiments, we compare the performance of individual features in terms of error and percentage error as summarized in Fig. 11. It can be seen that amongst the investigated features, spectral features report the lowest error values. EMD, fractal and statistical features report more or less similar performance but errors vary from one signal to another. For example, statistical features are able to model the Signals 5,6,8 and 10 with high accuracy but their performance deteriorates for other signals. On the other hand, raw signals and features based on Shannon envelop report high error values. Since Shannon envelop is directly derived from the signal, hence, errors due to raw signal and Shannon-based features are the highest among all the features. In general, the absolute error of spectral features varies between 5 and 10 bpm, for statistical features, it varies between 5 and 20 bpm, however, for raw signals, error increases beyond 20 bpm for some signals. Furthermore, we also show percentage contribution of each feature in the total error (Fig. 11-b). For instance, Signal 1 reports total error of 89.10 and out of this error the contribution from spectral features is 3.2 which maps to 2.70% of total error. It can be observed from the figure that the contribution of spectral features towards total error is the least among all the investigated features.

Finally, to provide further insights, we report the estimated heart rate as a function of time (for three different subjects) in Fig. 12. First two of these (Signal 13 and Signal 16) represent normal, while the third one (Signal 22), represents an abnormal subject. Test samples (shown in red) are distributed through out the signal to avoid any bias. While low errors are observed for Signals 16 and 22, a relatively high estimation error can be seen in case of Signal 13. Overall, the performance of the proposed method in terms of heart rate estimation is quite promising.

5.2. Subject-independent protocol

Most of the machine learning-based techniques reported in the

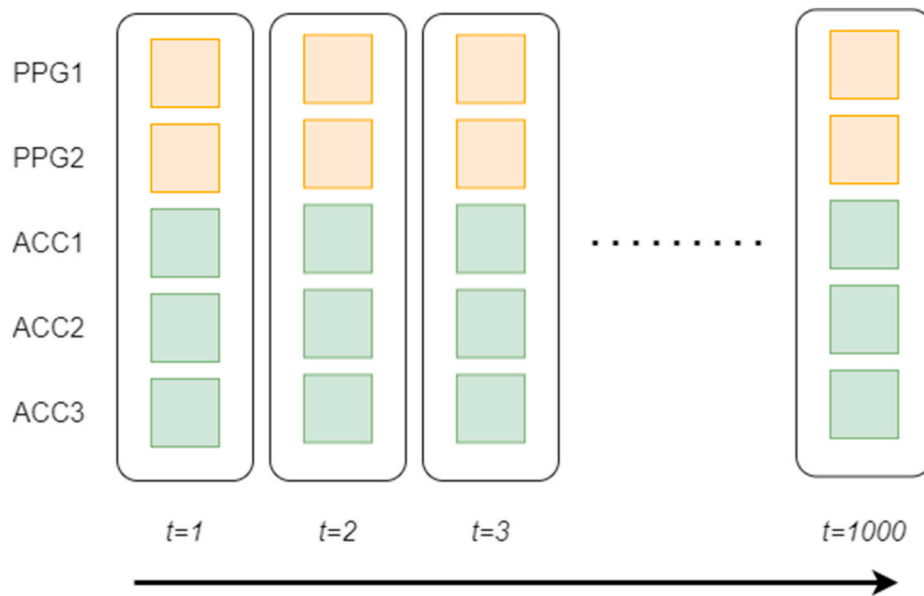


Fig. 10. Raw input window (5×1000) fed to the C-RNN model for learning the HR predictions.

Table 8

Performance of the C-RNN model fed with raw data and feature sequences. The features-based approach clearly outperforms the technique based on raw signals fed as features.

Subject	C-RNN with Raw Data		C-RNN with features	
	Error 1	Error 2	Error 1	Error 2
S1	6.79 ± 8.23	6.35 ± 7.95	3.02 ± 4.56	2.45 ± 3.91
S2	6.34 ± 8.52	9.97 ± 11.42	1.63 ± 1.46	1.45 ± 1.41
S3	4.57 ± 4.84	3.60 ± 3.76	2.92 ± 3.25	2.33 ± 2.81
S4	7.16 ± 9.25	9.36 ± 13.96	1.49 ± 1.70	1.29 ± 1.86
S5	5.73 ± 11.27	6.53 ± 9.62	3.04 ± 2.69	2.47 ± 4.70
S6	4.86 ± 11.45	3.59 ± 9.57	4.02 ± 7.23	5.16 ± 8.13
S7	3.98 ± 3.86	2.88 ± 2.60	0.90 ± 0.67	0.76 ± 0.65
S8	5.57 ± 7.64	4.60 ± 6.01	2.63 ± 3.34	2.32 ± 6.84
S9	4.12 ± 5.42	3.35 ± 4.40	2.25 ± 4.32	2.07 ± 4.11
S10	2.28 ± 2.90	1.52 ± 2.09	2.86 ± 3.21	1.46 ± 3.67
S11	3.53 ± 3.60	2.49 ± 2.68	1.65 ± 1.85	1.18 ± 1.38
S12	8.09 ± 8.53	6.22 ± 7.51	2.02 ± 1.15	1.47 ± 0.87
S13	5.01 ± 4.89	6.43 ± 5.50	3.80 ± 3.74	5.67 ± 5.17
S14	3.78 ± 2.94	5.04 ± 4.02	0.98 ± 1.48	1.21 ± 1.74
S15	8.77 ± 8.97	7.63 ± 8.68	2.38 ± 3.34	2.04 ± 2.61
S16	5.66 ± 5.24	3.76 ± 3.79	3.60 ± 4.67	2.51 ± 4.56
S17	6.83 ± 5.17	5.58 ± 4.15	4.40 ± 3.46	2.75 ± 2.91
S18	6.54 ± 4.95	4.91 ± 3.86	1.98 ± 2.89	1.51 ± 2.45
S19	6.26 ± 4.92	7.06 ± 5.69	1.40 ± 2.76	2.14 ± 3.53
S20	6.55 ± 8.55	4.48 ± 5.43	3.30 ± 4.12	1.97 ± 4.34
S21	4.88 ± 4.73	3.83 ± 3.71	1.85 ± 1.35	1.44 ± 1.02
S22	1.83 ± 2.03	2.11 ± 2.31	0.91 ± 0.61	1.05 ± 0.78
S1–S22	5.42 ± 6.27	5.06 ± 5.85	2.41 ± 2.90	2.12 ± 3.15

literature are evaluated in a subject-specific experimental protocol where a separate model is trained for each subject. In our study, in addition to the subject-specific modeling, we also carried out evaluations in the more challenging subject-independent protocol where a single model is trained using data from multiple subjects. These experiments are aimed at assessing the robustness of the proposed technique in the challenging experimental scenario. We carry out four different experiments as summarized in Table 7. Similar to subject-dependent experiments, we evaluate the system using raw signal values as well as using features for training the model. For the raw signals ($R(T_1, T_2)$), a relatively high average error of 9.2 ± 10.15 bpm is reported. Naturally, the error reduces significantly (3.8 ± 2.3) when feeding features to the model ($F(T_1, T_2)$) and the trend is similar to subject-dependent experiments. It is important to recall that these two experiments employ

signals from Type-01 exercises in the training set which are mainly whole body exercises like jogging etc. The test data of 10 signals involves Type-02 exercises which are primarily the upper body exercises where fists and arms are mainly used. This high variance in the training and test sets makes learning quite a challenging task. Despite the challenging scenario, the reported results are indeed quite promising. Likewise, when using a subset of samples from the 10 Type-02 signals as part of training ($F(T_1 + T'_2, T_2 - T'_2)$) and mixing the two types of exercises and randomly distributing the samples into training and test sets ($F(M)$), further reduces the estimation errors. This reduction can be attributed to the fact that once both Type-01 and Type-02 signals are mixed and then distributed into training and test sets, it allows an enhanced representation of the variation in signals in the training set which eventually results in better performance on the test set. A performance comparison of different experiments in terms of Pearson's correlation and Bland Altman plots can be seen in Fig. 13 and Fig. 14. Furthermore, we also present a comparison of average error (in bpm) for the four subject-independent experiments ($Raw(T_1, T_2)$, $F(T_1, T_2)$, $F(T_1 + T'_2, T_2 - T'_2)$, $F(M)$) and the subject-specific (SS) experiments (Fig. 14-e). Naturally, the least average error is reported in subject-specific experiments, however, the performance of subject-independent experiments (especially for the last two scenarios) is also comparable indicating the generalization of the proposed technique.

6. Discussion

We now present a performance comparison of the proposed model with recent studies on this problem in Table 9. The table includes both signal processing and machine learning-based techniques. It can be seen from Table 9 that among machine learning-based methods, the proposed technique outperforms all the listed studies except CorNET [20] and DeepHeart [54]. It is however, important to mention that DeepHeart [54] is evaluated only on the 12 signals in the training set which correspond to Type-01 exercises only. Likewise, although the error rates reported by CorNET [20] are slightly lower as compared to those reported by our model, it must be noted that CorNET is evaluated using a leave-one-out experimental protocol i.e. one of the windows is employed in the test set and all other windows are used in the training set. We, on the other hand, split the data into training (70%), validation (10%) and test (20%) sets which relates more closely to the real world scenarios. Despite a more challenging experimental protocol, a comparable mean

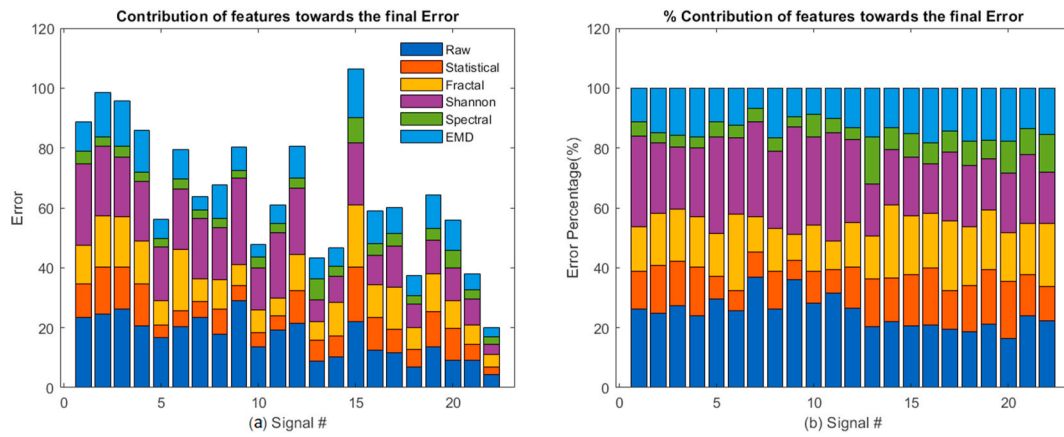


Fig. 11. Performance of individual features. (a): Absolute error reported by each feature. (b): Percentage contribution of each feature towards the final error. The least error is contributed by spectral features while raw signals contribute major share of the total error.

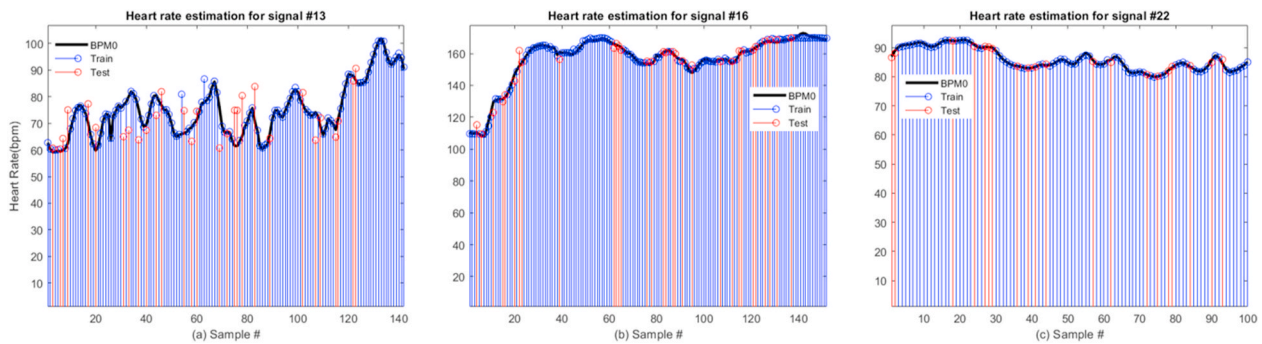


Fig. 12. Estimated heart rates as a function of time (a,b): Normal subjects (c): Abnormal subject. For all three subjects, blue samples represent training and red samples refer to inference data. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

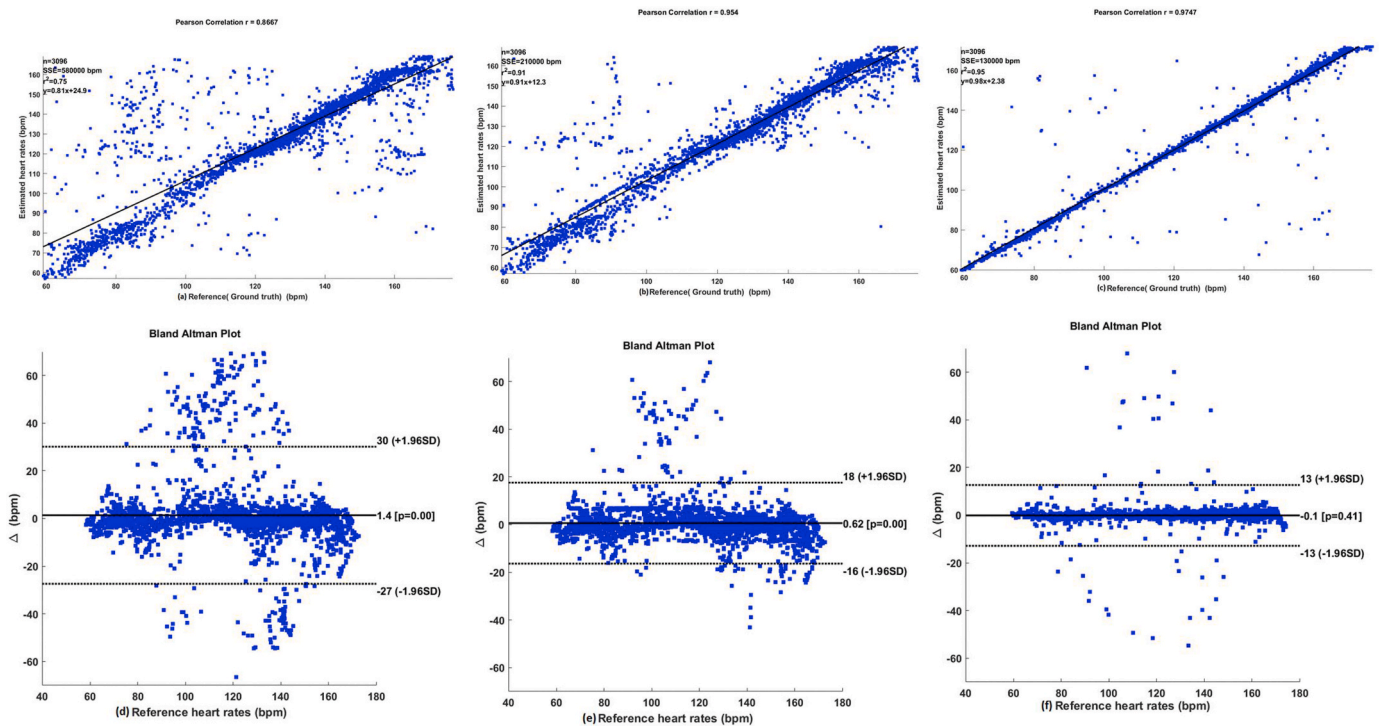


Fig. 13. Pearson's correlation and Bland Altman plots for the subject-independent experiments in the terms of beats per minute (bpm). (a) Pearson's correlation plot for the reference and the estimated heart rates employing the experimental setup $Raw(T_1, T_2)$. (b) Pearson's Correlation plot for the experimental setup $F(T_1, T_2)$. (c) Pearson's Correlation for $F(M)$ results in coefficient value of 0.9747. (d–f) Bland Altman plot for the three experimental protocols.

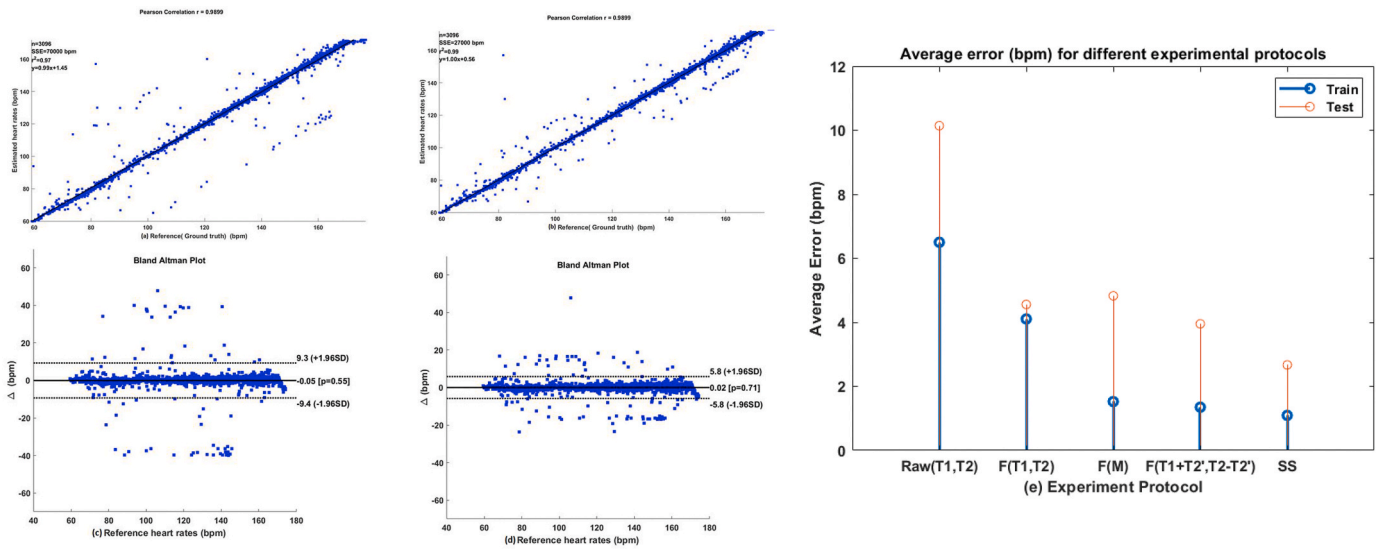


Fig. 14. Pearson's Correlation and Bland Altman plots for the subject-independent protocols in the terms of beats per minute (bpm). (a) Pearson's Correlation plot for the reference and the estimated heart rates employing experimental setup $F(T_1 + T'_2, T_2 - T'_2)$. (b) Pearson's Correlation plot for the subject-dependent protocol. (c) Bland Altman plot for the experiments $F(T_1 + T'_2, T_2 - T'_2)$. (d) Bland Altman plot for subject-specific (SS) protocol (e) Average errors for subject-independent and the subject-specific experiments. Error in the subject-dependent experiments is less than the subject-independent experiments and using features clearly outperforms the results with raw signals.

Table 9

Performance comparison with state-of-the-art. Mean absolute error along with the parameters are employed for the comparison.

Subject	Number of Signals	Parameters	Average Error
Troika [23]	22	–	2.73 ± 1.69
Joss [25]	22	–	2.15 ± 2.01
EEMD [42]	12	–	1.02 ± 1.79
WFPV [63]	22	–	1.97 ± 2.48
Chen et al. [41]	12	–	0.87 ± 1.28
CurToSS [31]	15	–	5.00 ± 2.8
CorNET [20]	22	256,578	1.47 ± 3.37
CorNET Mod. [20]	22	256,578	5.55 ± 4.45
b-CorNET [46]	22	–	6.7 ± 5.49
RTL [46]	22	–	7.32 ± 5.68
NAS-PPG [56]	15	0.8 M	6.0 ± 0.61
DeepHeart [54]	12	0.4 M	1.61
Risso et al. [57]	15	5-230k	3.84
Q-PPG [55]	15	269k	3.61
Proposed			
Subject-Specific	22	179,297	2.41 ± 2.90
Subject-Independent	22	179,297	3.80 ± 2.30

absolute error of 2.41 is indeed very promising. Among signal processing-based techniques, low error rates are reported by Chen et al. [41], Khan et al. [42] and Temko et al. [63]. A direct comparison of our technique with these methods is, however, not possible due to a different experimental protocol.

In addition to performance quantization using error, pre-processing, model parameters and generalization of model also play an important role in machine learning approaches. In our technique, we first process the signal using an Elliptical filter followed by feature extraction. All features are directly related with local signal properties. In case of DeepHeart [54], the signal is heavily pre-processed using a DCNN. Similarly, for Q-PPG [55], NAS-PPG [56] and Risso et al. [57], signal is pre-processed using DNN, NNI and MorphNet, respectively. Aside from pre-processing, model parameters represent an important factor during training and hardware implementation. The total number of trainable parameters in our model is 179,297 which is significantly less than 256,578 parameters in the CorNet [20] and 0.8 and 0.4 Million parameters in NAS-PPG [56] and DeepHeart [54]. Reduction of parameters not only

allows fast convergence during training, it also results in lesser memory and battery requirements at the inference time hence, allowing such systems to be deployed in wearable devices. Moreover, fewer parameters are likely to allow development of less expensive yet, more compact devices on which PPG sensor can be mounted. In addition to the number of parameters, few studies also report the delay between feeding the input and generation of output (i.e. execution time). In most cases, the execution time is reported as the average time to process a segment of the signal. The segment length is mostly kept constant and the most typical value employed in majority of the studies is a segment of length 8 s. Few studies also employ latency as a measure of complexity. However, execution times are a function of the hardware on which the system is implemented hence directly comparing these for different studies is not meaningful. Number of parameters, on the other hand, allows a more objective comparison of different techniques independent of the hardware platform.

In addition to comparison of overall mean absolute error, we also present a signal-wise comparison of our model with well-known machine learning-based techniques. This comparison is summarized in Table 10. The most closely related work to our solution is the CorNET system and its variants as they rely on deep learning-based frameworks and are using the same dataset as that of our experiments. As mentioned earlier, our method reports the lowest error rate among all the listed studies with the exception of CorNet [20]. We, however, employ a more stringent experimental protocol as opposed to the one employed in Ref. [20]. Other recent studies employing the SPC dataset include the works by Risso et al. [57], Q-PPG [55] and NAS-PPG [56]. These methods, however, report results on a subset of the dataset and our technique outperforms all of these studies reporting the least error on the complete set of signals in the dataset.

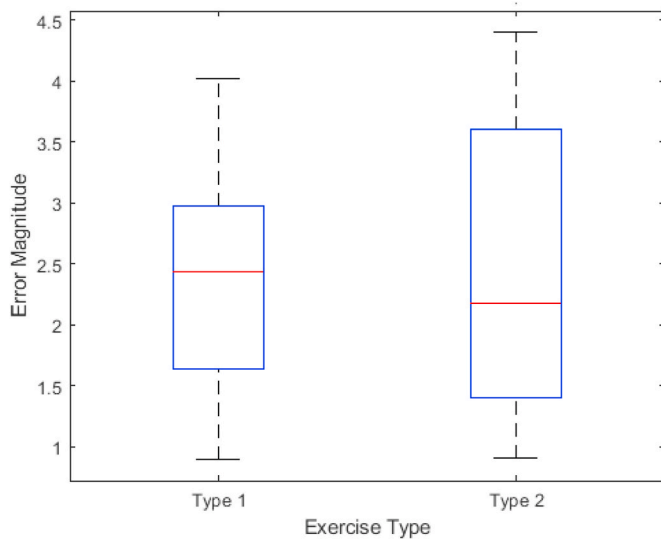
We also separately present the errors for whole body and upper body exercises in Fig. 15. It can be observed from the figure that the system reports a similar performance on both Type-01 and Type-02 exercises. This validates the generalization of the proposed model that is able to encompass both types of motion without a significant deviation in performance.

From the discussion above, it is clear that developed technique is generalized and robust. Not only the reported error rates are comparable to (and in most cases better than) the current state-of-the-art, the

Table 10

Performance comparison with state-of-the-art using machine learning approaches. Error used as metric in the listed studies is the Mean Absolute Error.

Subject	CorNET [20]	CorNET Mod. [20]	b-CorNET [46]	RTL [46]	NAS-PPG [56]	Risso et al. [57]	Q-PPG [55]	Proposed
S1	6.23 ± 9.44	4.26 ± 3.01	5.70 ± 4.14	5.80 ± 4.07	5.4 ± 8.11	3.17	3.25	3.02 ± 4.56
S2	1.83 ± 5.18	9.10 ± 7.52	9.51 ± 7.89	9.38 ± 9.19	5.0 ± 7.62	2.74	2.55	1.63 ± 1.46
S3	0.89 ± 3.49	3.79 ± 3.31	5.52 ± 4.29	5.37 ± 3.24	3.7 ± 4.75	3.13	2.66	2.92 ± 3.25
S4	0.49 ± 2.29	3.22 ± 3.13	3.75 ± 2.96	4.48 ± 3.95	6.4 ± 8.06	4.25	4.21	1.49 ± 1.70
S5	0.40 ± 1.01	2.35 ± 1.82	3.18 ± 2.58	4.44 ± 2.70	12.6 ± 21.08	4.88	5.41	3.04 ± 2.69
S6	3.08 ± 6.47	2.76 ± 1.98	3.38 ± 3.30	4.20 ± 3.47	10.5 ± 19.71	3.7	4.11	4.02 ± 7.23
S7	1.34 ± 4.42	2.18 ± 1.36	2.49 ± 1.67	3.53 ± 2.76	3.3 ± 3.79	2.48	2.06	0.90 ± 0.67
S8	3.64 ± 10.19	10.84 ± 8.68	13.30 ± 11.77	15.00 ± 11.26	8.07 ± 11.1	5.19	5.07	2.63 ± 3.34
S9	3.30 ± 6.81	1.68 ± 1.04	3.17 ± 2.30	3.86 ± 2.90	7.91 ± 10.06	7.00	7.15	2.25 ± 4.32
S10	1.77 ± 3.96	8.49 ± 7.06	12.84 ± 9.24	13.12 ± 11.74	3.29 ± 7.51	3.47	3.04	2.86 ± 3.21
S11	0.41 ± 1.37	3.22 ± 2.19	7.42 ± 5.46	7.92 ± 4.74	7.05 ± 11.33	3.67	3.07	1.65 ± 1.85
S12	0.50 ± 1.05	4.14 ± 3.40	4.17 ± 3.75	4.22 ± 3.49	6.76 ± 10.6	3.91	3.39	2.02 ± 1.15
S13	1.60 ± 2.29	9.67 ± 8.75	11.19 ± 9.17	12.90 ± 12.22	3.84 ± 4.88	2.85	2.13	3.80 ± 3.74
S14	0.24 ± 0.56	4.35 ± 3.44	5.39 ± 4.67	6.66 ± 5.66	4.85 ± 7.48	3.55	3.13	0.98 ± 1.48
S15	1.60 ± 3.87	7.03 ± 6.96	8.56 ± 7.15	8.77 ± 7.89	3.57 ± 5.75	3.6	2.96	2.38 ± 3.34
S16	2.04 ± 5.02	12.62 ± 9.15	13.63 ± 11.53	13.22 ± 11.40	–	–	–	3.60 ± 4.67
S17	0.95 ± 3.02	5.41 ± 4.63	5.08 ± 3.52	5.92 ± 4.51	–	–	–	4.40 ± 3.46
S18	0.28 ± 0.60	1.98 ± 2.89	4.31 ± 2.79	3.16 ± 3.09	–	–	–	1.98 ± 2.89
S19	0.28 ± 0.65	5.28 ± 3.54	6.09 ± 5.45	8.15 ± 7.15	–	–	–	1.40 ± 2.76
S20	0.67 ± 1.09	11.45 ± 9.27	10.25 ± 9.49	11.83 ± 9.83	–	–	–	3.30 ± 4.12
S21	0.42 ± 0.73	3.67 ± 2.79	5.76 ± 5.24	5.79 ± 4.37	–	–	–	1.85 ± 1.35
S22	0.57 ± 0.80	2.35 ± 2.16	3.15 ± 2.12	2.92 ± 2.13	–	–	–	0.91 ± 0.61
S1–S22	1.47 ± 3.37	5.55 ± 4.45	6.67 ± 5.49	7.32 ± 5.68	6.0 ± 10.61	3.84	3.61	2.41 ± 2.90

**Fig. 15.** Error (in terms of mean and standard deviation) for Type-01 and Type 02 exercises. Error for both type of body motions (Whole body (Type 01) and Upper body (Type 02)) is comparable.

performance is quite promising in the challenging subject-independent experimental protocol which is rarely attempted by the existing machine learning-based techniques. The model is also efficient in terms of computational resources. Despite a very good overall performance by the proposed technique, heart rate estimation for few signals reports relatively large error. One such example is shown in Fig. 12-a where the changes in heart rate are more abrupt causing estimation errors during training as well as the inference phase. Furthermore, in the present solution, the stopping criteria for EMD is difficult to control especially for the signals contaminated with motion artifacts. The time for decomposition of signal using EMD varies from one signal to another depending upon its quality. As a result, feature generation for EMD takes more time for noisy signals. We therefore, plan to extend the work by introducing SSA decomposition rather than EMD as the former is more canonical and data independent. Another limitation of the subject-dependent models is that a separate model is to be trained for every subject. This, however, is a common limitation of all studies which employ a subject-dependent

protocol. Nevertheless, our technique is generalized in the sense that the same solution can be employed both for subject-dependent and subject-independent scenarios.

7. Conclusion

We presented a machine learning-based solution to estimate heart rate from PPG signals. The proposed model comprises a combination of 1D convolutions with long short-term memory layers in a regression framework to learn heart rate estimation given a signal window. The model is fed both with raw signals as well as with feature sequences extracted from the raw signals, the later being the more robust solution. Experimental study on a publicly available dataset of PPG signals validated the effectiveness of the proposed technique in terms of feature extraction, computation cost and error rates.

In our further study on this problem, we intend to extend the idea of heart rate estimation to heart rate tracking where abnormal jumps/drops in heart rate can be identified which can eventually serve as potential indicators of heart-related problems. Furthermore, heart rate estimation from single PPG channel can also be investigated which in turn would result in fewer parameters as well as reducing the complexity of the solution. Likewise, the analysis of PPG can be further enhanced by complementing it with information from other modalities.

Declaration

I, Shahid Ismail, on behalf of myself and co-authors testify that our work titled 'Heart rate estimation in PPG signals using Convolutional-Recurrent Regressor' is our own work. The presented research material is not in consideration for publication in part or as a whole elsewhere.

Author's contributions

Shahid Ismail is a PhD candidate at Bahria University, Islamabad, Pakistan. He carried out research on heart rate estimation in PPG signals as a part of his PhD research work. Imran Siddiqi, Principal Supervisor of the candidate, contributed to algorithmic development and paper writing. Usman Akram, co-supervisor of the candidate, also contributed to the technical as well as non-technical aspects of the paper. All authors read and approved the final manuscript.

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Declaration of competing interest

Authors declare that they have no identifiable conflict or competing interest.

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